INTERNSHIP TRAINING REPORT

Allison Transmission
India Pvt Ltd

Allison Transmission

Internship report submitted by

NAME: VIDHYA S

DEPT: MECHATRONICS ENGINEERING

CLG: BANNARI AMMAN INSTITUTE OF TECHNOLOGY

DURATION:

FROM 04/03/2026 TO 02/04/2026

Transmission System (Automobile)

DEFINITION:

A transmission is a mechanical system that transmits power from the engine to the wheels and controls speed and torque.

Working Principle:

- Engine rotates torque converter
- Torque is transferred to transmission
- Planetary gears change speed/torque
- Clutches engage to select gear
- Power is sent to wheels

Types of Transmission:

- Manual Transmission
- Automatic Transmission

Purpose of Transmission:

- Transfers engine power to wheels
- Varies speed and torque
- Allows forward and reverse motion

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SECTION 1

Company Profile & Introduction

About Allison Transmission India Pvt Ltd

About Allison Transmission

Global Leader in Automatic Transmissions & Hybrid Propulsion

Founded

1915, headquartered in Indianapolis, Indiana, USA. One of the world's largest manufacturers of fully automatic transmissions.

Products

Fully automatic transmissions for buses, trucks, emergency vehicles, military, and off-highway equipment.

India Operations

Allison Transmission India Pvt Ltd, Oragadam, SIPCOT Industrial Park, Sriperumbudur, Chennai – manufacturing hub for gear components.

Core values & Vision

Customer Focus, Quality, Integrity, Innovation, Teamwork

Leading global propulsion solutions for a better working world

Global Presence

Over 100 years of innovation. Products used in more than 150 countries across commercial vehicles, defense, and energy sectors.

Oragadam Plant – Overview

Chennai, Tamil Nadu | Oragadam Special Economic Zone

Location

Oragadam, SIPCOT Industrial Park,, Sriperumbudur, Chennai – Tamil Nadu's automobile hub

Manufacturing Focus

Precision gear components for automatic transmissions and Manufactures 1000 & 2000 Series, 3000 Series, 4000 Series transmissions.

Production Lines

Fabrication Line | Housing Line | Assembly Line

Quality Standards

IATF 16949 - Quality Management System
ISO 14001 - Environmental Management System
ISO 45001 – Health & Safety

Workforce

Skilled engineers, operators & technicians across multiple shifts

Export

Components exported globally as part of Allison's worldwide supply chain

Organizational Structure & Departments

Key Departments at Oragadam Plant

Production

Ring, Green, Heat Treatment , Hard line , Housing & Assembly lines

Quality

In-process & final inspection, CMM,
Roll Checker - Gear meshing quality
PD Gauge - Gear size accuracy
Gear Analyzer - Full gear inspection
Hardness Test - Material strength

Maintenance

Preventive, predictive & TPM-based autonomous maintenance

Engineering

Process engineering, tool design & CNC programming

Stores & Logistics

Raw material, WIP & finished goods management

SHE

Safety , Health & Environment
Ergonomic workplace design
Health Management
HIRA
LOTO

HR & Administration

Workforce management, training & Education, Employee Engagement, Employee Engagement

Safety

Personal protective Equipment
Hazard Identification & Risk Assessment (HIRA)
Material Handling Safety
Fire Safety System

SECTION 2

Plant Layout & Production Lines

Production Lines at a Glance

Fabrication Line

- Internal ring gear manufacturing using broaching. Critical for torque transfer in the planetary gear set.
- Produces pinion gears that mesh with ring and sun gears. Key processes: hobbing, shaping, broaching, Heat treatment process and deburring.
- Maintains dimensional consistency before entering machining stage.

Housing Line

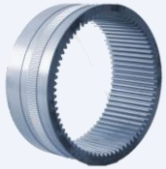
- Machining of transmission housings using CNC Machines(VMC/HMC).
- Ensures high dimensional accuracy for proper assembly fit.
- Performs operations like milling, boring, drilling, and tapping.
- Maintains tight tolerances and surface finish requirements.
- Uses precision inspection tools (CMM, gauges) for quality verification.

Assembly Line

- Final assembly of transmission sub-assemblies. Incorporates all machined components with quality checks at each station.
- Assembly of transmission components like gears, shafts, and clutch packs.
- Ensures correct fitment and alignment of all parts.
- Applies specified torque during fastening operations.
- Performs final testing such as leak test and performance test before dispatch.

Process Flow Details

Allison Transmission India Pvt Ltd



Ring

Broaching → Hobbing → Shaping → Deburring → Nitriding → Hard turning

Sun

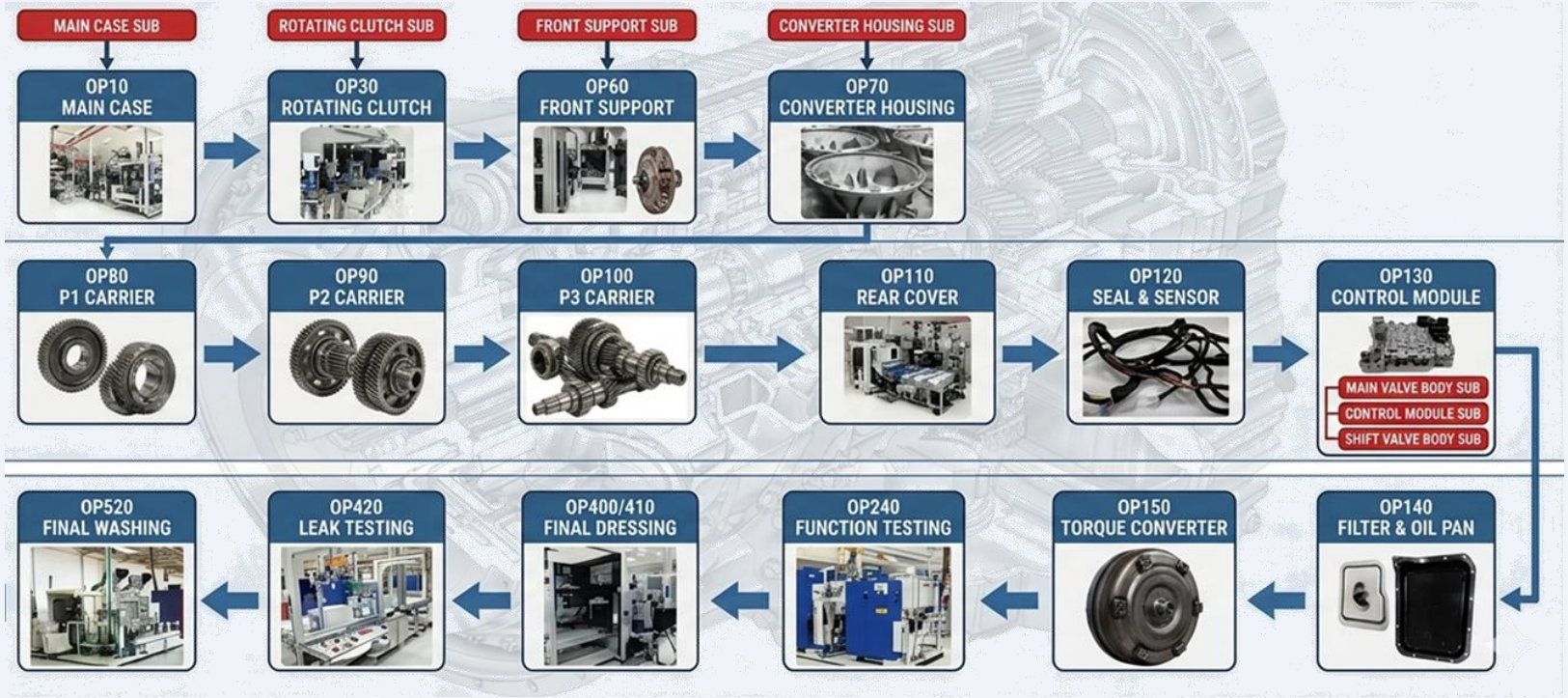
Hobbing → Shaving & chamfering → Shaping → Carburizing → Turning → *Intra* → *Laser welding*

Pinion

Hobbing → Shaving & chamfering → Carburizing → Face Grinding → Honing → Burnishing → *Intra*

Assembly Process Flow

Allison Transmission India Pvt Ltd



SECTION 3

Manufacturing Processes

Hobbing | Shaving | Shaping | Broaching | Heat Treatment

Gear Manufacturing Processes

DEFINITION

A gear is a mechanical component with teeth that mesh with another gear to transmit power and motion between machine parts.

GEAR MANUFACTURING PROCESSES

Broaching

Broach tool removes material progressively in a single pass to form gear teeth. Suitable for internal gears.

Hobbing

A rotating hob tool continuously cuts gear. Suitable for external gears.

Gear Manufacturing Process

Shaping

A reciprocating cutter cuts gear teeth. Suitable for internal and external gears
Advantages: Can cut complex gears and Good precision.

Shaving

Finishing process after cutting and removes small material from gear teeth.
Advantages: Improves surface finish and prevent Rough surface.

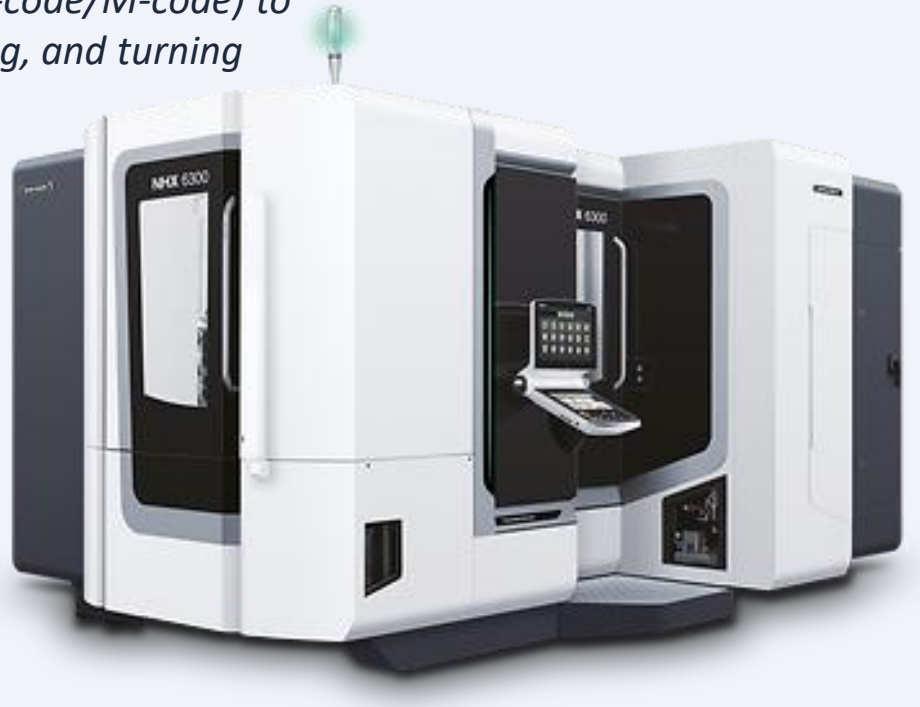
Heat Treatment

Heating and cooling the gear to improve properties. 3 types namely Hardening, Carburizing and Tempering.
Advantages: Increases strength and Improves wear resistance.

CNC Operations

CNC Machine (Computer Numerical Control)

A machine tool controlled by a computer program (G-code/M-code) to perform precise operations like cutting, drilling, milling, and turning automatically.



Working Principle:

- Program is written (G-code)
- Controller reads instructions
- Motors move tool/workpiece accurately
- Operation is performed with high precision

Advantages:

- High accuracy and repeatability
- Complex shapes can be made easily

CNC Turning Operations

Primary material removal process across all production lines

CNC turning forms the foundation of all gear blank preparation at Allison. Multi-axis CNC lathes perform OD turning, ID boring, facing, grooving and chamfering in single-setup operations to minimize errors and cycle time.

OD Turning	**ID Boring**	**Face Milling**	**Grooving**
Thread Turning	**Chamfering**	**Drilling**	**Tapping**

Prevents:

- Human errors in Machining.
- Dimensional inaccuracies
- Material wastage
- Poor surface finish

Frequency (Maintenance Check):

- Daily: Clean machine, check lubrication, coolant level
- Weekly: Inspect tool wear and alignment
- Monthly: Check calibration and electrical system
- During shutdown: Full servicing and software check

SECTION 4

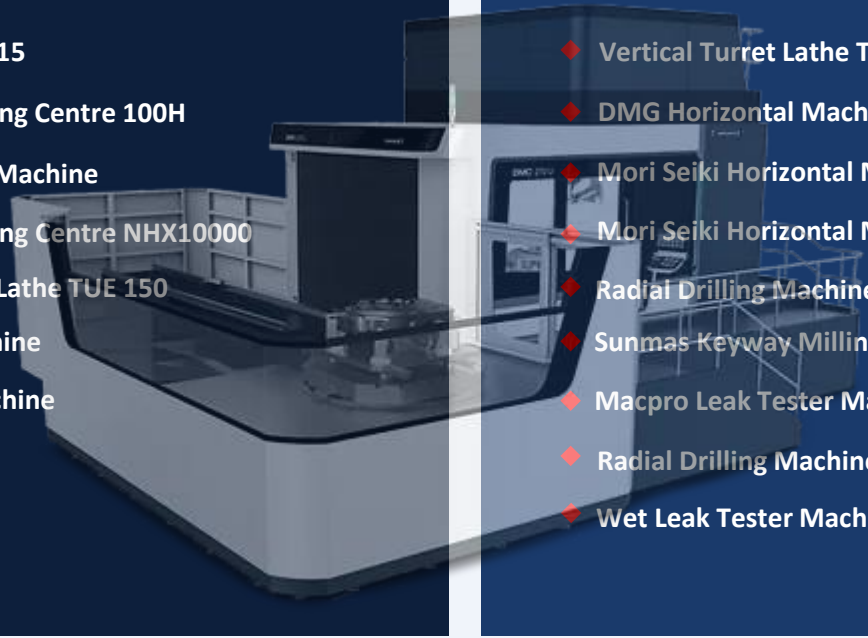
Machines & Equipment

Gleason | DMG Mori | Mitsubishi – World-Class Machining

Gear Line Machine

Gleason GP 300 Hobbing	MITSUIBISHI Shaping	NAGEL Honing	REDIN Deburring
Gleason 130 H – Hobbing & shaving	FORST Broaching	Surface Combustion Carburizing	NITREX Nitriding
C & B Machinery grinding	EMAG Turning	Jenfab Washing	Intra Gear Checker

Housing Line Cell – 1 & 2



HOUSING LINE CELL - 1

- Vertical Turret Lathe TU 15
- DMG Horizontal Machining Centre 100H
- Sunmas Keyway Milling Machine
- DMG Horizontal Machining Centre NHX10000
- Shibaura Vertical Turret Lathe TUE 150
- Macpro Leak Tester Machine
- Macpro Pre-Washine Machine
- Tool presetter
- Tool Crib

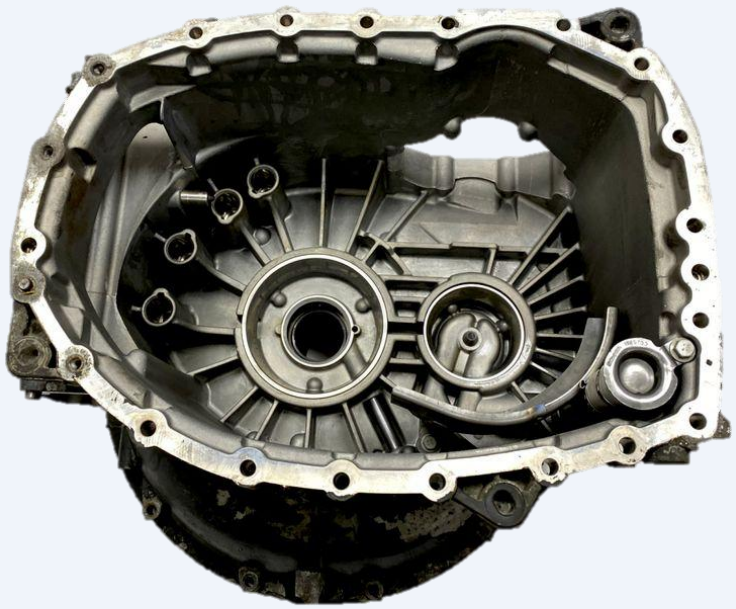
HOUSING LINE CELL - 2

- Vertical Turret Lathe TU 150
- DMG Horizontal Machining Centre 100H
- Mori Seiki Horizontal Machining Centre 6300
- Mori Seiki Horizontal Machining Centre 8000
- Radial Drilling Machine RM 65
- Sunmas Keyway Milling Machine
- Macpro Leak Tester Machine
- Radial Drilling Machine
- Wet Leak Tester Machine

Housing Line Cell - 3

HOUSING LINE CELL - 3

- Vertical Turret Lathe TU 15
- DMG Horizontal Machining Centre 80H
- DMG Mori Horizontal Machining Centre 6300
- Macpro Leak Tester Machine
- Macpro Pre-Washine Machine
- Wet Leak Tester (CMM LAB)
- Radial Drilling Machine
- Fixture Storage Rack
- Macpro Final Washing
- Housing Line Firewall



Housing Line

Quality Control Equipment

CMM lab (Coordinate Measuring Machine)

3D dimensional verification of machined components. Checks bores, profiles, GD&T features with high accuracy.

Gleason Gear Tester

Automated gear measurement for profile, lead, pitch error and surface finish. SPC integration for process control.

Roundness Tester

Checks circularity and cylindricity of critical bores and journals on gear components.

Surface Roughness Tester (Profilometer)

Ra/Rz measurement on tooth flanks, bore surfaces and sealing faces. Portable and benchtop versions.

Hardness Tester (Rockwell/Vickers)

Post-heat treatment hardness verification. HRC for case hardening depth and core hardness.

Quality standard

IATF 16949 - Quality Management System
ISO 14001 - Environmental Management System
ISO 45001 – Health & Safety

SECTION 5

TPM & Autonomous Maintenance (Jishu Hozen)

Total Productive Maintenance – TPM Award Preparation

Total Productive Maintenance – TPM Overview

Allison's commitment to world-class plant maintenance

Allison Transmission India Pvt Ltd is currently under TPM Award preparation. TPM aims for zero breakdowns, zero defects, and zero accidents through structured pillar activities.

AM **Autonomous Maintenance**	**PM** **Planned Maintenance**	**QM** **Quality Maintenance**	**FI** **Focused Improvement**	**EEM** **Early Equipment Management**	**ET** **Education & Training**	**SHE** **Safety Health Environment**	**OM** **Office TPM**

★ **Jishu Hozen (Autonomous Maintenance) is the first and foundational TPM pillar – the internship focus area**

Jishu Hozen – Autonomous Maintenance

Jishu Hozen is the practice of empowering machine operators to perform basic maintenance activities independently, detect abnormalities early, and keep equipment in peak condition. It shifts the mindset: "I operate, you maintain" → "I own my machine."

Step 1	**Initial Cleaning & Inspection**	Deep clean equipment; identify and tag abnormalities (fuguai)
Step 2	**Eliminate Contamination Sources**	Address sources of dirt, leakage, hard-to-clean areas
Step 3	**Initial Maintenance Standards**	Develop CIL(Cleaning, Inspection, Lubrication) standards
Step 4	**General Inspection**	Operator training on machine mechanisms; expand skills
Step 5	**Autonomous Inspection**	Operators conduct scheduled inspections per OPL/SOP
Step 6	**Standardization**	Standardize workplace – 5S integration, visual management
Step 7	**Full Autonomous Maintenance**	Operator-led maintenance; continuous improvement

Fuguai Tagging & Abnormality Elimination

Fuguai means abnormality or defect in machine condition. During Jishu Hozen Step 1, operators systematically identify all abnormalities using colored tags, log them, and track elimination progress.

BLUE TAG

Type: Operator-fixable

Loose bolts, minor leaks, cleaning issues, missing guards

RED TAG

Type: Maintenance-required

Electrical issues, complex mechanical failures, oil leaks from seals

FUGUAI ELIMINATION PROCESS

Identify

Operator spots abnormality during cleaning/inspection

Tag

Attach red/blue fuguai tag to the exact location

Log

Entry in fuguai register with date, machine, description

Resolve

Fix within target date; operator or maintenance team

Close

Tag removed; entry closed in register; verified

One Point Lesson (OPL)

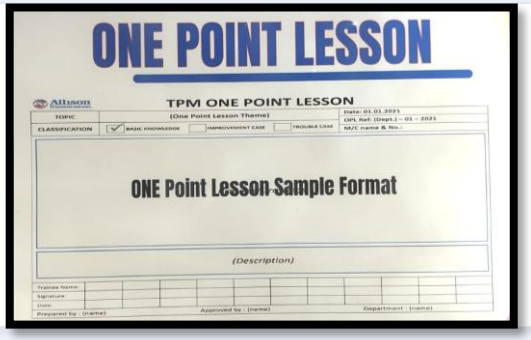
Simple visual knowledge-sharing tool in Jishu Hozen

One Point Lessons (OPL) are single-page, visual documents created to share knowledge on a specific topic. They enable operators to learn quickly at the machine level without lengthy training sessions. Allison uses OPLs extensively as part of JH implementation.

Basic Knowledge OPL	Improvement Case OPL	Abnormality Case OPL
Explains how a machine part works (e.g., how a hydraulic clamp functions). Builds operator understanding.	Documents a before/after improvement made to the machine or process. Prevents recurrence and shares best practices.	Describes a breakdown or defect that occurred, root cause, and corrective action. Used to avoid repetition.

OPL Format & Usage at Allison

Each OPL includes : Machine name | Date | Category | Topic description with diagrams/photos | Before & After status | Training sign-off column. Posted at the machine and reviewed during daily TPM meetings.



CLRI Standards – Cleaning, Inspection, Lubrication & Re-Tightening

CLRI (Cleaning, Inspection, Lubrication, Re-Tightening) Standards define the what, where, how, and how often of basic machine care. They are visual documents prepared by operators in collaboration with maintenance teams and form the basis of autonomous maintenance routines.

C CLEANING

- Remove chips, coolant residue, and contamination daily
- Use designated tools – brushes, air guns, cloth
- Prevents: wear acceleration, defects from contamination
- Visual standard: 'This is how it should look'

I INSPECTION

- Check bolts, guards, seals, oil levels during cleaning
- Identify early warning signs before failure occurs
- Use fuguai tags for abnormalities found
- Frequency: Daily, weekly, and monthly inspection points

L LUBRICATION

- Apply right lubricant at right point in right quantity
- Lubrication maps created for each machine
- Prevents: premature bearing/gear failure, increased wear
- Frequency: Shift-wise, daily, weekly as per schedule

R Re-Tightening

- Tighten the specified Nut, bold, Screws, etc.
- Using the mentioned technique up to the specified standard.
- Prevents: Loosening of components due to vibration, Equipment failure / breakdown.
- Frequency: Daily, weekly, and monthly inspection points

7 Types of Abnormalities

1. Minor Flaws (Defects) - Small, hidden issues like slight looseness, damaged parts, or minor corrosion that can escalate.

2.Unfulfilled Basic Conditions - Failure to maintain the fundamental operating standards, such as improper cleaning, lubrication, or tightening.

3. Inaccessible Places - Areas that are difficult to reach, hindering inspection, cleaning, or lubrication, leading to hidden failures.

4. Sources of Contamination - Areas where dust, oil leaks, or foreign materials (rust, chips) accumulate, affecting machine performance.

5. Sources of Quality Defects - Conditions that directly cause defects in the final product, such as misalignment or worn components.

6. Unnecessary Items - Items in the work area that are not required for current operations, such as extra tools, unused pipes, or trash.

7.Unsafe Places/Hazards - Environmental hazards, such as slick floors, missing guards, or sharp edges that pose a safety risk.

SECTION 6

Lean Manufacturing – 5S

Sort | Set | Shine | Standardize | Sustain

5S Implementation at Allison

Foundation of Lean Manufacturing & workplace discipline

5S is strictly followed across all production lines at Allison Transmission India. It creates a clean, organized, and standardized work environment that enables zero-defect production and supports TPM pillars.

1

SEIRI

SORT

Eliminate unnecessary items from the workplace. Red-tag strategy: items not needed in 30 days are removed. Reduces clutter and confusion.

2

SEITON

SET IN ORDER

A place for everything, everything in its place. Shadow boards, labeled tool storage, floor markings for material flow paths.

3

SEISO

SHINE

Clean the workplace daily. Cleaning is inspection. Find abnormalities during cleaning. Machines, floors, and aisles maintained immaculately.

4

SEIKETSU

STANDARDIZE

Create visual standards for the first 3S. SOPs, checklists, color coding systems. Best practice captured and shared across lines.

5

SHITSUKE

SUSTAIN

Discipline to maintain the first 4S. Daily audits, 5S score cards, management walkthroughs, operator accountability boards.

Other Lean Tools Deployed

Value Stream Mapping (VSM)

Maps the entire manufacturing flow to identify wastes (muda). Used to reduce lead time and WIP inventory between operations on gear lines.

Poka-Yoke (Error Proofing)

Fixtures, jigs, sensors, and limit switches prevent incorrect setup or loading. Prevents passing defective gears to next operation.

SMED (Quick Changeover)

Reduces setup/changeover time on hobbing and shaving machines. Separates internal and external activities; targets < 10 min changeover.

Visual Management

Color-coded floor markings, kanban cards, production status boards (Andon), and machine health indicator lights across all lines.

Kanban / Pull System

WIP between operations controlled through kanban signals. Prevents overproduction and manages inventory to takt time requirements.

Root Cause Analysis (Why-Why / Fishbone)

Root Cause Analysis is a technique used to find the real cause of a problem using methods like Why-Why analysis and Fishbone diagram.

SECTION 7

Six Sigma & Kaizen

Data-Driven Improvement & Continuous Innovation

Six Sigma at Allison

DMAIC methodology for defect elimination

Allison Transmission India follows Six Sigma principles to eliminate process variation and reduce defects to near-zero levels. The DMAIC (Define–Measure–Analyze–Improve–Control) methodology is applied to critical quality issues.

DEFINE

Define the problem, scope, CTQ (Critical to Quality) parameters, and project charter. Identify customer requirements.

MEASURE

Collect baseline data on process performance. Establish current sigma level (Cpk), defect rates, and measurement system analysis (MSA/GR&R).

ANALYZE

Root cause analysis using Ishikawa diagrams, Pareto analysis, regression, and hypothesis testing to identify vital few causes.

IMPROVE

Implement and test solutions. DOE (Design of Experiments) used to optimize cutting parameters for gear quality improvement.

CONTROL

Sustain improvements using control charts (SPC), updated control plans, revised SOPs, and ongoing monitoring dashboards.

Kaizen – Continuous Improvement Culture

Small daily improvements that create big results

'Kaizen' means change for the better. Allison cultivates a strong kaizen culture where every employee – operators, engineers, and managers – actively contributes improvement ideas. Kaizens are raised, reviewed, implemented, and recognized formally.

Quick Kaizen (QK)

Small, quick-win improvements that can be implemented within a day by the operator. No CAPEX required. Examples: repositioning tools, adding labels, shadow boards.

Standard Kaizen

Structured improvement requiring cross-functional team. Typically completed in 1–2 weeks. Involves root cause analysis and verification.

Major Kaizen / Kaikaku

Radical improvement or breakthrough change. Requires management approval, budget. Examples: line reconfiguration, new tooling strategy.

SECTION 8

Internship Work & Observations

My Contribution during the Internship Period

Week-wise Internship Activities

Week 1	Plant orientation, safety induction, SHE briefing, production line walkthrough & introduction to TPM framework Study of manufacturing processes: hobbing, shaving, shaping, broaching on sun gear & pinion lines
Week 2	CLRI Standards, Fuguai tagging and Focus on TPM activities. Learning about six sigma and Understanding tools in CNC machining
Week 3	Study of Lean Manufacturing Process: 5s, Kaizen, Value Stream Mapping, TPM, Kanban, Poka-yoke. Housing Line: CNC Machine Operations and CNC Programming (G-Code & M-Code).
Week 4	Assembly process of transmission and Testing (leak test, functional test) Final report preparation and Submission

Key Observations on Shop Floor

Safety

PPE compliance strictly enforced. Safety tags on all energy isolation points. Incident-reporting culture is positive and non-punitive.

5S Discipline

Exceptional cleanliness and organization observed across sun gear and pinion lines. Shadow boards 100% complete. Floor markings crisp and visible. Daily 5S patrol visible.

Machine Utilization

OEE boards displayed at each machine. Downtime logged and analyzed. Breakdown response teams deployed within minutes of machine stoppage.

Quality Mindset

100% in-process inspection on critical dimensions. Operators trained to use air gauges and profilometers. Rejection bins are labeled and tracked.

TPM Culture

Operators actively participate in fuguai tagging, CLRI activities, and OPL review. TPM activity board updated daily. Strong operator ownership observed.

SECTION 9

Key Learnings & Skills Developed

Technical, Managerial & Professional Growth

Technical Knowledge Gained

Gear Manufacturing Processes

Deep understanding of hobbing, shaving, shaping, broaching, turning – process parameters, tooling, quality outcomes and machine types.

Machine Familiarity

Hands-on exposure to Gleason, DMG Mori, and Mitsubishi CNC machines; understanding of CNC control interfaces and setup sheets.

Metrology & Inspection

Learned to use gear testers, CMM, gauges, roughness testers, and interpret measurement data against drawing specifications.

TPM – Autonomous Maintenance

Practical experience in Autonomous Maintenance, fuguai tagging, CIL standard development, and OPL creation.

Lean & Quality Tools

Applied knowledge of 5S, Kaizen, DMAIC, VSM, and visual management techniques in a live production environment.

Failure Analysis

Participated in breakdown analysis using Why-Why and Ishikawa diagram methods; learned FMEA structure.

Professional & Soft Skills Developed

01 Team Collaboration

Worked alongside operators, engineers, and TPM team members. Learned to communicate technical observations clearly.

02 Adaptability

Adapted quickly to the strict, safety protocols, and structured routines of a world-class automotive manufacturing plant.

03 Time Management

Managed internship tasks, audits, and documentation within shift schedules in a fast-paced manufacturing environment.

04 Analytical Thinking

Used data and observation to identify abnormalities, propose kaizen ideas, and contribute to improvement activities

05 Observation Skills

To identify shop floor abnormalities – key skill for both quality and TPM activities.

Theory vs. Real-World Practice

Bridging classroom knowledge with industry experience

Topic	*Classroom Theory*	*Industry practice at Allison*
Gear Manufacturing	Formulae for module, pressure angle, addendum	Actual Gleason hobbing setups, SPC-controlled processes
Machine Tools	Diagrams of lathe, milling machine components	DMG Mori NLX multi-axis CNC with live tooling
Quality Control	Cp, Cpk definitions and calculations	Real-time SPC dashboards; CMM reports on production parts
Maintenance	Preventive maintenance concepts	JH steps, CLRI standards, fuguai tags in live production
Lean / 5S	Textbook 5S definitions	5S audits, shadow boards, kaizen ,6 Sigma reviews on shop floor
Safety	SHE theory	PPE enforcement, LOTO procedures, daily safety briefings

SECTION 10

Conclusion & Acknowledgement

Summary, Takeaways & Gratitude

Conclusion

Summary of the internship experience

This internship at Allison Transmission India Pvt Ltd, Oragadam provided an exceptional opportunity to experience world-class automotive precision manufacturing first-hand. Working under the Autonomous Maintenance (Jishu Hozen) pillar of TPM offered deep practical insight into how global manufacturers achieve and sustain excellence.

Process Mastery

Gained comprehensive understanding of gear manufacturing processes – hobbing, shaving, shaping, broaching and CNC operations.

TPM in Action

Witnessed in TPM; the discipline and systematic approach to the machine proactive maintenance.

Lean is a Culture

5S, Kaizen, Six Sigma practices at Allison. Every employee contributes to improvement daily.

Industry Readiness

The internship significantly enhanced technical competency, professional discipline, and readiness for a career in manufacturing engineering.

ACKNOWLEDGEMENT

I extend my sincere gratitude to all those who made this internship a truly enriching and memorable experience.

Allison Transmission India Pvt Ltd

For providing this internship opportunity in a world-class manufacturing environment and exposing me to best-in-class TPM and lean manufacturing practices.

TPM Coordinators / Mentor

For patiently guiding about each pillars, kaizen activities and providing constructive feedback throughout the internship.

Shop Floor Operators & Engineers

For their patience in demonstrating machine operations and sharing practical knowledge.

THANK YOU

Internship Report – Allison Transmission India Pvt Ltd, Oragadam

Gear Manufacturing | Autonomous Maintenance (Jishu Hozen) | TPM | Lean Manufacturing